CrewAI Multi-Agent Evaluation

The CrewAI implementation used three specialized agents: a Product Research Specialist, a Product Pricing and Comparison Analyst, and a Business Recommendation Writer. The sequential crew followed a fixed Research → Analysis → Writing pipeline, while the hierarchical crew used a Manager Agent to coordinate the specialist agents dynamically.

The sequential approach was more suitable for this product recommendation task because the workflow had clear dependencies and a predictable order. The first sequential run used 348 total tokens across 3 successful requests, while the hierarchical run used 51,723 total tokens across 37 successful requests. A later sequential run used 3,105 tokens across 9 successful requests. Exact monetary cost was not calculated because the model was accessed through the Hugging Face router and provider pricing was not fixed for this experiment, so token usage was used as the cost-awareness measure.

Manual Evaluation

Run	Factual Grounding	Completeness	Clarity & Tone	Average
Sequential Run 1	4/5	5/5	4/5	4.3/5
Hierarchical Run	3/5	5/5	4/5	4.0/5
Sequential Run 3	1/5	2/5	4/5	2.3/5

The first sequential run was mostly accurate, although the final writer introduced a few unsupported descriptive claims. The hierarchical run successfully demonstrated delegation, but it introduced an incorrect price comparison and additional unsupported claims. The third sequential run exposed an important reliability problem: the agent produced a completely different product dataset and did not call the read_csv tool, so its final recommendation was not grounded in the actual products.csv file.

Sequential vs Hierarchical

Aspect	Sequential	Hierarchical
Quality	More consistent for this task	More variable
Token usage	Much lower	Much higher
Latency	Lower expected overhead	Higher due to manager coordination
Reliability	More predictable	More opportunities for delegation errors
Main advantage	Simple and controlled workflow	Dynamic delegation and coordination
Main disadvantage	Fixed workflow	More complex and expensive
Best use	Predictable multi-step tasks	Complex tasks requiring dynamic delegation

Conclusion

For this specific product recommendation task, the multi-agent hierarchical crew was not worth the additional complexity and token usage compared with the simpler sequential workflow. The hierarchical run consumed substantially more tokens and also introduced a factual calculation error. The third sequential run additionally demonstrated that tool-based grounding can still fail when the agent does not actually invoke the required tool. Therefore, sequential execution or even a single well-designed agent would be preferable for this simple task, while hierarchical CrewAI would be more appropriate for complex projects that genuinely require dynamic delegation between specialists.